Subject: Mathematics | **WEEKLY CHAPTER TEST – 05** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 6: Triangles** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Marks are indicated against each question.
2. **Section A** consists of 5 questions of 1 mark each (4 MCQs and 1 Assertion-Reason).
3. **Section B** consists of 4 questions of 11 marks total (1 VSAQ of 2 marks and 3 SA of 3 marks each).
4. **Section C** consists of 1 Case-Based integrated unit of assessment of 4 marks with sub-parts.
5. Use of calculators is strictly prohibited. Neat geometric sketches must accompany geometric proofs.

SECTION A: MULTIPLE CHOICE QUESTIONS (1 Mark Each)

1. If two similar triangles have scale factor $a : b$, then the ratio of their corresponding altitudes is:

(a) $a^2 : b^2$ (b) $a : b$ (c) $\sqrt{a} : \sqrt{b}$ (d) $2a : b$

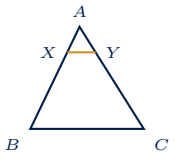
2. If two similar triangles have scale factor 3 : 4 and the perimeter of the smaller triangle is 36 cm, the perimeter of the larger triangle is:

(a) 48 cm (b) 45 cm (c) 54 cm (d) 42 cm

3. In ΔABC, D and E are midpoints of sides AB and AC respectively. Then the ratio of DE to BC is:

(a) 1 : 1 (b) 1 : 2 (c) 1 : 4 (d) 2 : 1

4. In the given figure, $XY \parallel BC$ in ΔABC. If $AX = 1$ cm, $XB = 3$ cm, and $BC = 6$ cm, then XY is:



(a) 1.5 cm (b) 2 cm (c) 1 cm (d) 2.5 cm

5. **Assertion (A):** If $\Delta ABC \sim \Delta DEF$ such that $\frac{AB}{DE} = \frac{2}{3}$ and perimeter of ΔABC is 16 cm, then perimeter of ΔDEF is 24 cm.
Reason (R): The ratio of the perimeters of two similar triangles is equal to the ratio of their corresponding sides.

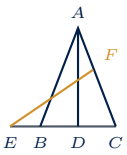
(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(c) Assertion (A) is true but Reason (R) is false.
(d) Assertion (A) is false but Reason (R) is true.

SECTION B: DESCRIPTIVE QUESTIONS (PART 1)

6. **[2 Marks]** If $\Delta ABC \sim \Delta DEF$ such that $AB = 5$ cm, $DE = 7.5$ cm, and the perimeter of ΔDEF is 36 cm, calculate the perimeter of ΔABC.

7. **[3 Marks]** Prove that the line segment joining the midpoints of any two sides of a triangle is parallel to the third side and equal to half of it.

SECTION B: DESCRIPTIVE QUESTIONS (PART 2 Continued)



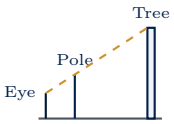
8. **[3 Marks]** In the given figure, E is a point on side CB produced of an isosceles triangle ABC with $AB = AC$. If $AD \perp BC$ and $EF \perp AC$, prove that:

$$\Delta ABD \sim \Delta ECF$$

9. **[3 Marks]** In ΔABC and ΔEFG, CD and GH are respectively the medians to sides AB and FE. If $\Delta ABC \sim \Delta FEG$, prove that: (i) $\Delta ADC \sim \Delta FHG$, (ii) $\frac{CD}{GH} = \frac{AC}{FG}$.

SECTION C: CASE STUDY BASED QUESTION (4 Marks)

10. **Case Study 05: Forestry & Tree Height Estimation Using Similar Triangles**



A forestry officer needs to measure the height of a mature redwood tree without climbing it. She places a vertical measuring pole of height 2.5 m at a distance of 36 m from the base of the tree. Walking back an additional 4 m along the line of sight, she aligns her eye with the top of the pole and the top of the tree. Her eye level is 1.5 m above the ground.

(i) **[1 Mark]** What is the height of the measuring pole above her eye level?

(ii) **[1 Mark]** What is the total horizontal distance from her eye to the base of the tree?

(iii) **[2 Marks]** Calculate the total vertical height of the redwood tree above the ground.

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